

Case Study Task

Request Management Application

Objective: Build a full-stack Request Management Web Application focusing mainly on the backend. The goal is to:

- Design and build a backend system that enforces real business rules.
- Write clean, maintainable, well-structured code.
- Show solving the requirements without over-engineering.

Requirements

- An employee (A) creates a request and assigns it to another employee (B).
- Employee (B) manager can approve/reject the request.
- Employee (B) can only action and close the request once it is approved by their manager.
- App must run locally

Backend (Main Focus)

- Set up a Backend application using PostgreSQL, Nodejs and Express
- Code Quality, Structure, Patterns, Maintainability
- REST APIs, Data Storage and Data Modelling.
- Business Logic & Validations
- Authentication & Authorization
- Design Patterns & Error Handling
- Logging, Metrics & Monitoring
- Two Roles: Employee & Manager

Frontend

- Build UI of your choice for Sign Up, Sign in, Create Request, Approve Request, Action & Close Request
- Crude UIs are fine. Purpose is to demonstrate how the full stack system works with the backend.

Submission

- Submit your GitHub repository link containing both frontend and backend folders.
- Share a live demo link (Vercel / Netlify, etc).